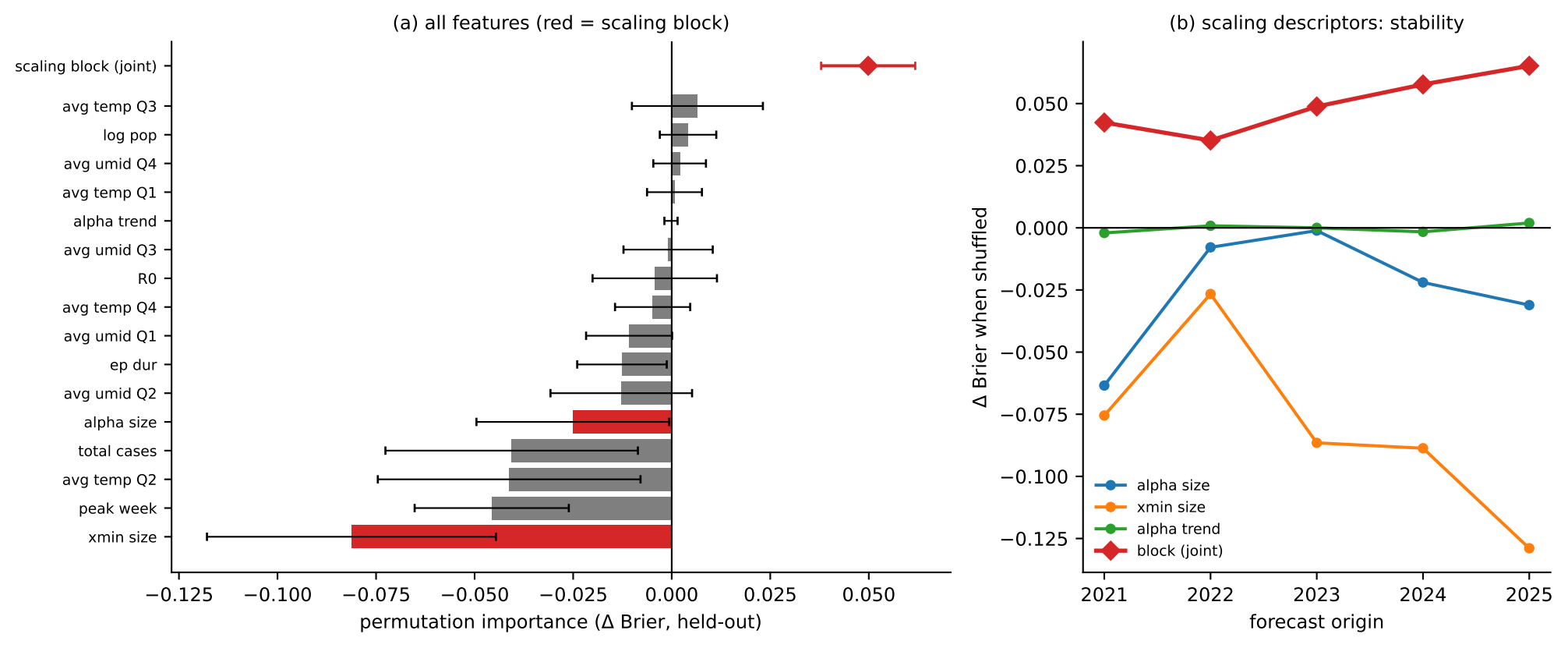
(a) all features (red = scaling block)
scaling block (joint)
avg temp Q3
log pop
avg umid Q4
avg temp Q1
alpha trend
avg umid Q3
R0
avg temp Q4
avg umid Q1
ep dur
avg umid Q2
alpha size
total cases
avg temp Q2
peak week
xmin size
−0.125
−0.100
−0.075
−0.050
−0.025
0.000
0.025
0.050
permutation importance (Δ Brier, held-out)
(b) scaling descriptors: stability
Δ Brier when shuffled
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
alpha size
xmin size
alpha trend
block (joint)
2021
2022
2023
2024
2025
forecast origin